



EXAMINATION PAPER

FACULTY : COMPUTER SCIENCE AND MULTIMEDIA
COURSE : BACHELOR OF INFORMATION TECHNOLOGY (Hons)
YEAR/ SEMESTER : THIRD YEAR / SEMESTER FIVE
MODULE TITLE : ARTIFICIAL INTELLIGENCE
CODE : BIT 353
TIME ALLOWED : 3 HOURS

Instruction to candidates

1. This question paper has THREE (3) Sections.
2. Answer **ALL** questions in Section A, MCQ.
3. Answer **5** questions in Section B, MSAQ.
4. Answer **2** questions in Section C, MEQ.
5. No scripts or answer sheets are to be taken out of the Examination Hall.
6. For Section A, answer in the OMR form provided.

Do not open this question paper until instructed

(Candidates are required to give their answers in their own words as far as practicable)

SECTION A
Multiple Choice Questions
Attempt All Questions

[30×1=30]

- 1. Who is considered to be the "father" of artificial intelligence?**
 - A. Fisher Ada
 - B. John McCarthy
 - C. Allen Newell
 - D. Alan Turing

- 2. The finding of a path start state to a goal state is known as:**
 - A. Searching
 - B. Classification
 - C. Regression
 - D. Simulation

- 3. A heuristic is a way of trying _____.**
 - A. to discover something or an idea embedded in a program
 - B. to search and measure how far a node in a search tree seems to be from a goal
 - C. to compare two nodes in a search tree to see if one is better than the other
 - D. All of the above

- 4. Consider a good system for the representation of knowledge in a particular domain. What property should it possess?**
 - A. Representational Adequacy
 - B. Inferential Adequacy
 - C. Inferential Efficiency
 - D. All of the Above

- 5. Prior to the invention of time-sharing, the prevalent method of computer access was _____.**
 - A. batch processing
 - B. telecommunication
 - C. remote access
 - D. all of the above

- 6. What is state space?**
- A. The whole problem
 - B. Your definition to a problem
 - C. Problem you design
 - D. Representing your problem with variable and parameter
- 7. A search algorithm takes _____ as an input and returns _____ as an output.**
- A. input, output
 - B. problem, solution
 - C. solution, problem
 - D. parameters, sequence of actions
- 8. Which value is assigned to alpha and beta in the alpha-beta pruning?**
- A. Alpha = max
 - B. Beta = min
 - C. Beta = max
 - D. Both Alpha = max & Beta = min
- 9. Which is created by using single propositional symbol?**
- A. Complex sentences
 - B. Atomic sentences
 - C. Composition sentences
 - D. None of the above
- 10. Which were built in such a way that humans had to supply the inputs and interpret the outputs?**
- A. Agents
 - B. AI system
 - C. Sensor
 - D. Actuators
- 11. The finding of a path start state to a goal state is known as:**
- A. Searching
 - B. Classification
 - C. Regression
 - D. Simulation

- 12. Which values are independent in minimax search algorithm?**
- A. Pruned leaves x and y
 - B. Every states are dependant
 - C. Root is independent
 - D. None of the mentioned
- 13. Which is the first AI programming language?**
- A. BASIC
 - B. FORTRAN
 - C. ILP (Inductive logic programming)
 - D. LISP
- 14. Which is more suitable normal form to be used with definite clause?**
- A. Positive literal
 - B. Negative literal
 - C. Generalized modus ponens
 - D. Neutral literal
- 15. What is state space?**
- A. The whole problem
 - B. Your Definition to a problem
 - C. Problem you design
 - D. Representing your problem with variable and parameter
- 16. Which is the best way to go for Game playing problem?**
- A. Linear approach
 - B. Heuristic approach (Some knowledge is stored)
 - C. Random approach
 - D. An Optimal approach
- 17. Values of alpha-beta search get updated _____.**
- A. at the end
 - B. along the path of search
 - C. initial state itself
 - D. none of the above

- 18.The area of AI that investigates methods of facilitating communication between people and computers is:**
- A. Natural Language Processing
 - B. Symbolic Processing
 - C. Decision Support
 - D. Robotics
- 19.How many possible sources of complexity are there in forward chaining?**
- A. 1
 - B. 2
 - C. 3
 - D. 4
- 20.The strategy used to reduce the number of tree branches and the number of static evaluations applied in case of a game tree is:**
- A. MinMax Strategy
 - B. Alpha Beta Pruning Techniques
 - C. Constrain Satisfaction Strategy
 - D. Static Max Strategy
- 21.What is Artificial Intelligent?**
- A. Putting your intelligence into Computer
 - B. Programming with your own intelligence
 - C. Making a Machine intelligent
 - D. Putting more memory into Computer
- 22.How can be the goal is thought of in backward chaining algorithm?**
- A. Queue
 - B. List
 - C. Vector
 - D. Stack
- 23.What form of negation does the prolog allows?**
- A. Negation as failure
 - B. Proposition
 - C. Substitution
 - D. Negation as success

24.What is the field of Natural Language Processing (NLP)?

- A. Computer Science
- B. Artificial Intelligence
- C. Linguistics
- D. All of the above

25.What is Machine Translation?

- A. Converts one human language to another
- B. Converts human language to machine language
- C. Converts any human language to English
- D. Converts Machine language to human language

26.An agent is composed of _____.

- A. architecture
- B. agent function
- C. perception sequence
- D. architecture and program

27.Which of the following is an advantage of using an expert system development tool?

- A. Imposed structure
- B. Knowledge engineering assistance
- C. Rapid prototyping
- D. All of the above

28.The game of Poker is a _____ agent.

- A. single
- B. double
- C. triple
- D. unknown

29.Which of the following is NOT a property of representation of knowledge?

- A. Representational Verification
- B. Representational Adequacy
- C. Inferential Adequacy
- D. Inferential Efficiency

30.What is the primary interactive method of communication used by humans?

A. Reading

B. Writing

C. Speaking

D. All of the above

SECTION B

Short Answers Questions

Answer any five (5) questions out of eight (8) questions [5×6=30]

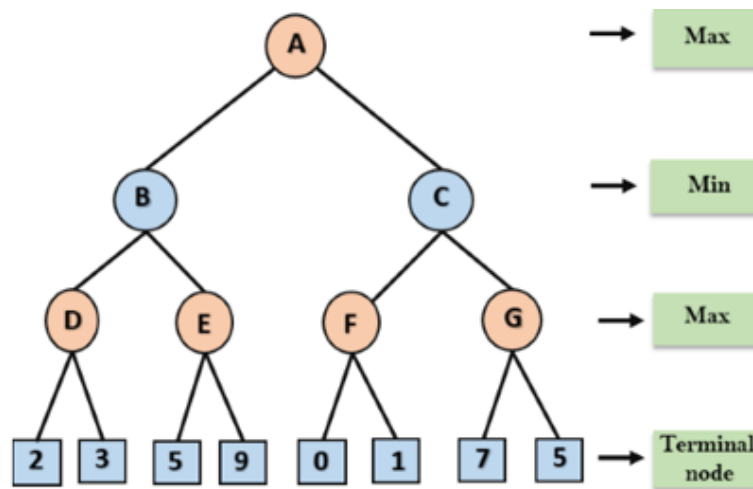
- 1. Define AI. Discuss how artificial intelligence will change the future. [2+4]**
- 2. What is Knowledge Representation? Enumerate the different approaches to represent the knowledge. [2+4]**
- 3. Write your understanding about expert system. Highlight the characteristics and advantages of an expert system. [2+4]**
- 4. What are Intelligent Agents and how are they used in AI? [2+4]**
- 5. Express the following knowledge using FOLP with quantifier.**
 - A. All man drinks coffee
 - B. Boy who is intelligent
 - C. Every man respects his parents
- 6. Define CSP. Outline some popular problems that can be solved by using CSP. [2+4]**
- 7. Show the difference between Forward and Backward Chaining.**
- 8. Define Turing Test and Total Turing test. Mention some common misconceptions about AI. [3+3]**

SECTION C

Long Answer Questions

Attempt any two (2) questions out of three (3) questions [2×20=40]

1. Shed light on how Machine learning works. Differentiate between supervised and unsupervised machine learning algorithm. [8+12]
2. Find the optimal move for the player using Alpha Beta pruning algorithm and also write the limitation of MINMAX. [15+5]



3. Define NLP. Classify and explain the steps for parsing NLP. Also, describe how it is using in the daily techno gadget devices with suitable example. [2+12+6]

******BEST OF LUCK******